

FINITE SOLUBLE GROUPS WITHOUT INCREASING UNREFINABLE CHAINS

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ABSTRACT. We answer negatively a question of Monakhov and Sokhor: must every finite group have an unrefinable subgroup chain with nondecreasing successive indices? For each odd prime p , we give a soluble subgroup of $\mathrm{GL}_5(p)$ of order $2^6 p^8$ with no such chain. The construction adapts an example of Kohler. Its maximal-subgroup indices force two downward steps of index p^2 , after which the only indices available are 2 and p^4 . Monotonicity would then force a subgroup of order divisible by p to be a 2-group.

1. INTRODUCTION

In a finite group G , an unrefinable subgroup chain

$$1 = G_0 < G_1 < \cdots < G_n = G$$

has G_{i-1} maximal in G_i for every i . We call it *increasing* if its successive indices $j_i = |G_i : G_{i-1}|$ satisfy $j_1 \leq \cdots \leq j_n$. Monakhov and Sokhor call this a $<$ -chain [2]. They show that a group with two maximal chains of the same length, one with nondecreasing indices and the other with nonincreasing indices, is supersoluble.

Monakhov and Sokhor asked the author whether every finite group has an increasing unrefinable chain. Finite nilpotent groups do: choose maximal chains in their Sylow subgroups and adjoin the Sylow factors in increasing order of the primes. Each step within a Sylow q -subgroup has index q . Solubility alone, however, does not suffice.

Theorem 1. *For every odd prime p , there is a soluble group $G_p \leq \mathrm{GL}_5(p)$ of order $2^6 p^8$ with no increasing unrefinable subgroup chain.*

The construction uses the $t = 2$ case of Kohler's example [1, Section 4, Theorem 4.1], which shows that bounds on maximal-subgroup indices need not pass to subgroups. We give a matrix realization and use its nested maximal subgroups to rule out increasing chains. The smallest member of the family is G_3 , of order 419904.

The reason for the obstruction is that a four-dimensional irreducible commutator layer lies in the Frattini subgroup of the normal p -subgroup. It therefore contributes no maximal index at the top. Removing either of two two-dimensional layers makes the commutator layer complemented, but its index p^4 is then too large to follow a downward step of index p^2 . The following proposition makes this obstruction precise.

2. THE INDEX OBSTRUCTION

Write $\mathcal{I}(A) = \{|A : B| : B \text{ is maximal in } A\}$ for the maximal-index spectrum of a finite group A . Spectra alone do not describe which maximal steps can follow one another; the conjugacy conditions below supply this information.

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Proposition 2. *Let p be an odd prime and let the finite group G have maximal subgroups M_1, M_2 with a common maximal subgroup N . Suppose that $p \mid |N|$ and*

$$\mathcal{I}(G) = \{2, p^2\}, \quad \mathcal{I}(M_1) = \mathcal{I}(M_2) = \{2, p^2, p^4\}, \quad \mathcal{I}(N) = \{2, p^4\}.$$

Suppose also that every maximal subgroup of G of index p^2 is conjugate to M_1 or M_2 , and every maximal subgroup of M_i of index p^2 is M_i -conjugate to N , for $i = 1, 2$. Then G has no increasing unrefinable subgroup chain.

Proof. Read a putative increasing chain downward, so that its indices must be nonincreasing. Once an index 2 occurs, all subsequent downward indices must equal 2, and the subgroup from which that step starts must be a 2-group.

Since p divides the orders of G and both M_i , the first two indices must therefore be p^2 . By the conjugacy assumptions, we may conjugate first in G and then in the chosen M_i to make these steps

$$G \xrightarrow{p^2} M_i \xrightarrow{p^2} N.$$

At N , the bound p^2 excludes the index p^4 , leaving only 2. This would make N a 2-group, contrary to $p \mid |N|$. $\square$

3. THE CONSTRUCTION

Fix an odd prime p , put $F = \mathbb{F}_p$, and let

$$s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D = \langle s, t \rangle \leq \mathrm{GL}_2(F).$$

Thus D is dihedral of order 8. With block sizes 2, 1, 2, set

$$L = \left\{ \ell(x, y, z) = \begin{pmatrix} I_2 & x & z \\ 0 & 1 & y \\ 0 & 0 & I_2 \end{pmatrix} : x \in F^{2 \times 1}, y \in F^{1 \times 2}, z \in F^{2 \times 2} \right\}.$$

The subgroup $H = \{\mathrm{diag}(A, 1, B) : A, B \in D\} \cong D \times D$ normalizes L , since conjugation $\ell \mapsto h\ell h^{-1}$ sends

$$(1) \quad (x, y, z) \mapsto (Ax, yB^{-1}, AzB^{-1}).$$

Define $G_p = L \rtimes H \leq \mathrm{GL}_5(p)$. It is soluble, being an extension of a p -group by a 2-group, and $|G_p| = p^8 \cdot 8^2 = 2^6 p^8$.

To identify the commutator subgroup and its module structure, put

$$X = \{\ell(x, 0, 0)\}, \quad Y = \{\ell(0, y, 0)\}, \quad Z = \{\ell(0, 0, z)\}.$$

Multiplication and commutation are given by

$$\begin{aligned} \ell(x, y, z)\ell(x', y', z') &= \ell(x + x', y + y', z + z' + xy'), \\ [\ell(x, y, z), \ell(x', y', z')] &= \ell(0, 0, xy' - x'y), \end{aligned}$$

where $[a, b] = a^{-1}b^{-1}ab$. In particular, Z is central in L , and $L' = Z$ because the outer products xy span $F^{2 \times 2}$. Also $\ell(x, y, z)^p = \ell(0, 0, pz + \binom{p}{2}xy) = 1$, since p is odd. The Frattini formula gives

$$(2) \quad \Phi(L) = L^p L' = Z.$$

As an H -module, $L/Z = X \oplus Y$, identifying each subgroup with its image. Both summands are irreducible of dimension 2: t fixes only the coordinate lines, and s interchanges them. Their distinct action kernels $1 \times D$ and $D \times 1$ show that they are nonisomorphic.

The action on Z in (1) is irreducible of dimension 4. Indeed, D spans the full matrix algebra $M_2(F)$: $(I_2 + t)/2$ and $(I_2 - t)/2$ are the diagonal matrix units, and multiplication by s gives the other two. An H -invariant subspace of $Z \cong M_2(F)$ is therefore stable under all left and right

matrix multiplications. Any nonzero matrix in such a subspace yields every matrix unit, so the subspace is all of Z . Thus the required irreducible dimensions are 2, 2, 4.

To compute the maximal indices, we use a standard consequence of Schur–Zassenhaus. If V is an elementary abelian p -group and H is a p' -group, the maximal subgroups of $V \rtimes H$ are $V \rtimes J$, with J maximal in H , and the V -conjugates of $W \rtimes H$, with W a maximal H -submodule of V . For completeness, a maximal subgroup T not containing V satisfies $VT = V \rtimes H$. Then $W = V \cap T$ is normal, and T/W is a complement to V/W , V/W -conjugate to WH/W by Schur–Zassenhaus. Maximality is exactly the irreducibility of V/W . In our case the first type has index 2, while a quotient of dimension d in the second type gives index p^d .

Lemma 3. *Set*

$$M_X = (Z \times X) \rtimes H, \quad M_Y = (Z \times Y) \rtimes H, \quad N = Z \rtimes H.$$

The subgroups M_X, M_Y are maximal in G_p of index p^2 , and N is maximal in each of them of index p^2 . Their spectra are

$$\mathcal{I}(G_p) = \{2, p^2\}, \quad \mathcal{I}(M_X) = \mathcal{I}(M_Y) = \{2, p^2, p^4\}, \quad \mathcal{I}(N) = \{2, p^4\}.$$

Every maximal subgroup of G_p of index p^2 is conjugate to M_X or M_Y . For each $M \in \{M_X, M_Y\}$, every maximal subgroup of M of index p^2 is M -conjugate to N .

Proof. First, every maximal subgroup M of G_p contains Z . Otherwise $G_p = ZM$, whence

$$L = Z(L \cap M) = \Phi(L)(L \cap M).$$

The defining property of the Frattini subgroup then gives $L \cap M = L$, contradicting $Z \not\leq M$.

It follows that the maximal subgroups of G_p are the inverse images of those of $G_p/Z = (X \oplus Y) \rtimes H$. Since X, Y are nonisomorphic irreducible modules, the only maximal H -submodules of $X \oplus Y$ are X and Y . The semidirect-product description above gives the spectrum $\{2, p^2\}$ and the two stated conjugacy types.

In M_X , the normal subgroup $Z \times X$ is elementary abelian, with nonisomorphic irreducible summands of dimensions 4 and 2. Its maximal H -submodules are Z and X , giving indices p^2 and p^4 ; index 2 comes from H . The index- p^2 subgroups are precisely the M_X -conjugates of $Z \rtimes H = N$. The same argument applies to M_Y . Finally, the irreducibility of Z gives $\mathcal{I}(N) = \{2, p^4\}$. $\square$

Proof of Theorem 1. The group G_p has the stated order and is soluble. Lemma 3 and $|N| = 2^6 p^4$ give all the hypotheses of Proposition 2. $\square$

4. THE EXAMPLE $p = 3$

The construction gives a compact generating set for G_3 . Over $\mathbb{F}_3$, write $e_{ij} = I_5 + E_{ij}$, where E_{ij} is a matrix unit. Then

$$(3) \quad G_3 = \langle e_{13}, e_{23}, e_{34}, e_{35}, \text{diag}(s, 1, I_2), \text{diag}(t, 1, I_2), \\ \text{diag}(I_2, 1, s), \text{diag}(I_2, 1, t) \rangle.$$

Indeed, the first four generators give X and Y , and their commutators generate Z . The relevant orders and spectra, with the preceding subgroups specialized to $p = 3$, are

A	$ A $	$\mathcal{I}(A)$
G_3	419904	$\{2, 9\}$
M_X, M_Y	46656	$\{2, 9, 81\}$
N	5184	$\{2, 81\}$.

The standalone GAP file `G3-verification.g` gives the eight generators as literal matrices, checks the orders, maximal inclusions and relevant conjugacy classes, and exhaustively excludes

increasing chains. The file and its output are available in the verification archive [3]. These calculations check the explicit example; the proof for every odd p is independent of computation and classification.

The construction illustrates how a Frattini layer can conceal a large maximal index until smaller indices have already been forced. For this purpose, a few maximal-index spectra together with their conjugacy data suffice; the full subgroup lattice is unnecessary. It remains to determine the least order of a counterexample. No minimality of G_3 beyond this family is asserted.

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